

EE 271 Final Project User's Manual: Frogger

Welcome to Frogger! This system implements the classic Frogger (or Crossy Road) game on a DE1-SoC FPGA board on a 16x16 Bi-Color LED expansion board. The player controls a frog (green pixel) on the LED Board Using the 4 KEYS on the DE1_SoC proto board. The goal is to get to the other side whilst avoiding the moving cars (red pixels). Winning fills the board green and displays "WIN" for WIN, whilst losing fills the board red and displays "LOSE" for LOSE.

Keybinds:

- Switch 9 (SW[9]):
 - This functions as a reset. Flipping the switch up "pauses" the game, and flipping the switch back down restarts the game.
- KEY[3]:
 - Moves the frog up (top of board).
- KEY[2]:
 - Moves the frog down (bottom of board).
- KEY[1]:
 - Moves the frog left.
- KEY[0]:
 - Moves the frog right.

Here's how the game works:

1. Hit the reset button to restart the game.
2. Use the keys to navigate to the other side of the board, whilst trying to avoid the cars.
3. If your frog makes contact with a car or a car makes contact with the frog, the game ends and you lose. If your frog makes it to the other side without colliding with any cars, you win!
4. Flip switch 9 up and back down to restart the game.

Happy Playing and Good Luck!